Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
5	(a)	(i)		Addition of water (1) Galactose and glucose correctly drawn (OH below C^1 on galactose + CH_2OH at C^6 on glucose) (1)	1	1		2		
		(ii)		Hydrolysis (1)	1			1		
		(iii)		Add Benedict's reagent/solution and heat (1) NOT ref to HCl Change of colour <u>from blue</u> to {yellow/green/orange/brown/(brick) red} (1)	2			2		2
	(b)	(i)		Glucose oxidase has a <u>specific</u> <u>shaped</u> active site (1) Only glucose molecules have a complementary (shape) (1) (Therefore glucose fits) to form an enzyme-substrate complex (1)	1	2		3		
		(ii)		Any **two** (x1) from: • More glucose the more hydrogen peroxide is produced (1) • Equation shows a one to one relationship / is directly proportional (1) • Concentration of hydrogen peroxide is converted to an electrical signal/ detected by the electrode (1)			2	2		

Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
	*(*c)			Any **four** (x1) from: A. Arabinose has a similar {shape/ structure} to glucose/ complementary to active site of glucose oxidase (1) Reject same shape B. So would act as a competitive inhibitor (of glucose oxidase)/ {competes for/ binds to} the active site (1) C. preventing the formation of enzyme-substrate complexes (1) D. Therefore, less hydrogen peroxide produced (1) E. The display would give a lower reading (1) Answer including description of non-competitive inhibition negates mark points A-C			4	4		
				Question 5 total	**5**	**3**	**6**	**14**	**0**	**2**